

Caden Cheng

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EDUCATION

University of California, San Diego

La Jolla, CA

B.S. in Computer Science and Mathematics, Data Science Minor

Expected Grad 2028

GPA: 3.53 | **Relevant Coursework:** Data Structures, Discrete Math, Systems Programming, Algorithms

EXPERIENCE

Software Engineer Intern

Jun. 2026 – Present

Barobo

San Jose, CA

- Stopped unauthorized video sharing by engineering single-use **AWS CloudFront** signed URLs with TTLs
- Scaled to **10,000+** monthly video requests by tracking user metadata in **AWS DynamoDB** for authorization
- Cut environment provisioning from 2 hours to 15 mins by writing reusable **Terraform** and **AWS CLI** scripts
- Decreased API response time by **40%** as measured by AWS CloudWatch, by **troubleshooting** and optimizing database queries

Software Developer → Incoming VP of Technology & Engineering Manager

Oct. 2025 – Present

Computer Science Engineering Society (CSES) at UCSD

La Jolla, CA

- Onboarded **100 users** across 10 labs, driving **web application development** for **Next.js** auth and profile flows
- Secured API access for **4 user roles** by building an **RBAC** layer with **10 scoped permissions** in **TypeScript**
- Hardened auth reliability by writing **11 Jest unit tests** covering getSession() permission and status checks
- Restricted **10 labs'** data to verified @ucsd.edu users by integrating **Google OAuth** with **NextAuth** and **JWT**
- Modeled user and role data by designing a **Mongoose** User schema with email validation and indexed lab info

PROJECTS

inVISION [\[Link\]](#) | *Python, OpenCV, MediaPipe, NumPy, Pygame*

Apr. 2026

- Pitched as an effective communicator to win **Best Interactive AI** at ACM Diamond Hacks (**500+ peers**)
- Achieved real-time 3D rendering at **60 FPS** by building a custom vision pipeline and interactive viewer
- Automated the conversion of 2D images into 3D geometries by integrating the **TripoSR** framework
- Built an optimized 3D rendering pipeline using **NumPy** for transformations, culling, and lighting internally
- Implemented a computer vision control scheme using **MediaPipe** for native pinch-to-zoom and rotation tracking

Silent Speech [\[Link\]](#) | *PyTorch, OpenCV, TensorFlow, Python*

Jan. 2026

- Won **#1 Best Overall Hack** at San D Hacks 2026 out of 35+ competing undergraduate teams
- Developed a real-time lip-reading AI using a **BiGRU**, achieving **85% accuracy** on live webcam feeds
- Engineered a pipeline using **OpenCV** for facial landmark extraction, normalizing features for scale invariance
- Optimized model training using stratified sampling, mixup, and noise injection data augmentation

remark. [\[Link\]](#) | *Python, TypeScript, React, Claude API, Deepgram, FastAPI*

Jun. 2026

- Shipped a hands-free AI voice agent in **24h** at **Berkeley AI Hackathon 2026**, owning audio + HUD
- Routed **95%** of spoken intents correctly across **10+ task types** by wiring a **3-tier** agentic router
- Cut speech-to-action latency to **2.5s** by building a real-time **React** HUD streaming captions and reasoning
- Protected user privacy by coding an on-device **OpenCV** face-gate that limits passive listening to active talk

ACM Projects Spotify Song Recommender | *Pandas, PyTorch, NumPy, Matplotlib*

Jan. 2026 – Apr. 2026

- Predicted user preferences with **85% accuracy** by developing a collaborative filtering **machine learning** model
- Processed and structured a massive dataset of **7.7M+ interactions** spanning 930K+ users and 30K+ songs
- Reduced model training time by **20%** by building a PyTorch DataLoader pipeline with negative sampling
- Optimized weights using Adam and MSE loss, evaluating generalization via Precision, Recall, and ROC AUC

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, C/C++, C#, HTML/CSS, SQL

Frameworks: React, Next.js, Node.js, Tailwind CSS, PyTorch, TensorFlow

Developer Tools: Git/GitHub, AWS (CloudFront, DynamoDB, CLI), Terraform, MongoDB Atlas, npm, Linux/Unix

Libraries: Pandas, NumPy, Matplotlib, Torchvision, OpenCV, MediaPipe, Mongoose